

Mini-Project Specification

Data-Driven Security Analysis and Defensive Response

APPARATUS

For this mini-project, apparatus means the tools, files, and working environment you need to complete the assessment.

You will need:

- a computer with Python available
- a terminal or command prompt
- a text editor or IDE
- access to an AI service if your group chooses to use one
- access to the course Moodle page
- your assigned authentication-log extract
- a working folder for your report, code, outputs, and supporting files

PURPOSE

The mini-project asks you to work in groups to analyse authentication-log evidence and use that evidence to reason about cyber security risk and defensive response. In Part 1, your group analyses an assigned extract from an `auth.log` dataset using Python. You produce reproducible evidence, visualisations, an asset-focused risk matrix, and an initial response recommendation. In Part 2, your group builds on one prioritised risk or finding from Part 1 and develops a technical defensive response.

The assessment emphasises evidence, reproducibility, interpretation, risk prioritisation, and clear communication.

Submission planning fields

Use this space to plan your group submission.

Group name or number

Group members

Assigned extract

Chosen asset

Part 1 evidence files

Part 2 selected risk or finding

Assessment structure and weighting

The mini-project is worth 20% of the overall course mark. The assessment has two connected parts.

Part	Weight	Summary
Part 1: Data-Driven Authentication Log Analysis	10%	Analyse an assigned authentication-log extract using Python and produce a short evidence-based report section, supporting code, visualisations, and an asset-focused risk matrix.
Part 2: Technical Defensive Response	10%	Build on one prioritised risk or finding from Part 1 and develop a technical defensive response.

There will be one final submission deadline for the mini-project. The two parts should be submitted together as one group submission package.

Learning outcomes assessed

The mini-project assesses the following Intended Learning Outcomes.

- **ILO 3:** identify and analyse common cyber threats and vulnerabilities affecting networks, web applications, and endpoints.
- **ILO 4:** demonstrate an understanding of secure system and software development principles through the design of simple, defensible architectures.
- **ILO 6:** understand the nature of countermeasures against forensic analysis, including common anti-forensic techniques and their implications for digital investigations.
- **ILO 7:** communicate cyber security risks, assumptions, and recommendations clearly and appropriately to both technical and non-technical audiences.

Scenario and dataset allocation

You are working as part of a small security analysis team asked to review authentication activity from a Linux-based system. The organisation has provided authentication log data and wants to understand whether the evidence suggests suspicious activity, which risks should be prioritised, and what defensive response would be appropriate.

Each group will be assigned an authentication-log extract. The extracts are taken from the same base `auth.log` dataset. The assessment task and marking criteria are the same for all groups, but the specific evidence, counts, source IP addresses, usernames, and time patterns may differ depending on the assigned extract.

Your analysis, Python script, visualisations, risk matrix, and recommendations must be based on your group's assigned `auth.log` extract. Do not assume that another group's findings apply to your own extract.

Part 1: Data-Driven Authentication Log Analysis

In Part 1, your group must analyse the assigned authentication-log extract. Your task is to identify suspicious or security-relevant authentication activity, support your findings with reproducible evidence, and communicate what the organisation should prioritise.

KEY PRINCIPLE

Your analysis should address this question:

Which authentication events or patterns should be prioritised for investigation or response, and why?

You may use AI tools to support or perform aspects of the log analysis. For example, AI may be used to help interpret the log format, suggest analysis questions, or identify possible patterns. However, AI-generated analysis is not sufficient evidence on its own. Any quantitative findings, patterns, tables, or visualisations used in your report must be verified using your own Python script against your assigned log extract.

You must write a Python script that verifies the evidence used in your report. The script should read the raw log file, extract relevant authentication information, generate summary outputs, and produce the visualisations used in your report. The script should be written as a reusable analysis artefact. A subsequent review team should be able to rerun it, understand how the evidence was produced, and adapt it to comparable authentication logs.

Your README and code comments should explain the key decisions behind your analysis, including how you identify failed authentication attempts, extract usernames and source IP addresses, handle unmatched or ambiguous log lines, and generate the outputs used in your report.

Your analysis should include:

- a focused security question or framing
- Python-verified findings from the assigned log extract
- at least two Python-generated visualisations
- an asset-focused risk matrix containing 3 to 5 risks or findings
- initial response recommendations
- discussion of assumptions, uncertainty, missing evidence, and limitations

Part 1 submission requirements

Report content. Your Part 1 work should be included within the single mini-project report. The overall report must not exceed **6 pages**. The title page, references, and appendices do not count towards this overall page limit.

The Part 1 section of the report should present your security question or framing, explain your Python-verified findings, interpret the evidence, summarise the risks, and recommend an initial response.

The Part 1 section should include or refer to at least two Python-generated visualisations. One visualisation must show the top source IP addresses by failed authentication attempts. At least one further visualisation should be chosen by your group to support your analysis, such as targeted usernames, failed attempts over time, event-type distribution, privileged-account attempts, or another justified view of the data.

Appendices to the report. The report appendices should include:

- an asset-focused risk matrix containing 3 to 5 risks or findings

- an AI-use declaration, if AI tools were used
- a one-page workload record, if the group wishes to include one

If no AI-use declaration is included, the course team will assume that the group is declaring that no AI tools were used. If no workload record is included, the course team will assume that the group is declaring that all members contributed equally.

Python and reproducibility files. Submit the following files alongside the report:

- README.md
- analysis.py
- an output folder containing the tables, figures, and other files generated by your script

The README should explain how to run the script, the required Python version and packages, the expected input file, the generated outputs, and any assumptions or limitations in the parsing or analysis.

Log file

Each group must use the authentication log file assigned in Table 1. Groups should not analyse or compare against logs assigned to other groups.

Authentication log file	Assigned groups
1_auth.log	Group A, Group AA, Group AB, Group AD, Group AE, Group AF, Group B, Group BA, Group BB, Group BC
2_auth.log	Group BF, Group BI, Group BJ, Group BK, Group BL, Group BM, Group BO, Group C, Group D, Group E
3_auth.log	Group F, Group G, Group H, Group I, Group J, Group K, Group L, Group N, Group O, Group P
4_auth.log	Group Q, Group R, Group S, Group T, Group V, Group W, Group X, Group Y, Group Z

Table 1: Authentication log allocation by group. Each group must analyse only its assigned authentication log file.

Asset-focused risk matrix

Your risk matrix should be tied to a clearly identified asset. Likelihood and impact should be assessed in relation to that asset, not in the abstract. If your analysis concerns more than one asset, you should make clear which asset each risk assessment applies to.

Do not treat a raw event count, such as a high number of failed logins, as a risk by itself. The risk is the possible consequence for the asset, such as attempted unauthorised access, reduced availability of authentication, weakened confidence in access control, or the need for further investigation.

CHECKPOINT

Before completing the risk matrix, check that you can answer:

- What is the asset?
- Why does this asset matter to the organisation?
- What could happen to this asset?
- What evidence supports the risk judgement?
- What uncertainty remains?

Part 2: Technical Defensive Response

In Part 2, your group will build on one prioritised risk or finding from Part 1. You should select a risk or finding from your risk matrix and develop a technical defensive response that would help the organisation prevent, detect, reduce, or respond to that risk.

The technical response may take different forms, depending on the nature of the risk selected. Possible directions include:

- detection logic for suspicious authentication behaviour
- logging and alerting improvements
- rate-limiting, logout, or access-control measures
- a simple defensible architecture or configuration change
- another technically justified defensive response

Your Part 2 response should explain:

- which Part 1 risk or finding you are addressing
- why this risk or finding is a suitable technical priority
- what defensive response you propose
- how the response would reduce likelihood, reduce impact, improve detection, or support investigation
- what limitations, trade-offs, or residual risks remain

Whole submission package

There will be one final group submission for the mini-project. The submission should contain both Part 1 and Part 2 materials.

Your written submission should be a single report. The report must not exceed **6 pages in total**, with **no more than 3 pages for Part 1** and **no more than 3 pages for Part 2**. The title page, references, and appendices do not count towards this page limit.

The main report should present the key evidence, interpretation, risk prioritisation, recommendations, and technical defensive response. Appendices may be used for supporting material such as the asset-focused risk matrix, AI-use declaration, workload record, supplementary tables, and additional figures. However,

the main argument should be understandable from the 6-page report. Do not use appendices to hide essential analysis, justification, or recommendations.

A suggested submission structure is shown below.

FILE STRUCTURE

```
mini-project-submission/  
  report.pdf  
  part1/  
    README.md  
    analysis.py  
    output/  
      summary_counts.csv  
      top_source_ips.csv  
      top_source_ips.png  
      chosen_visualisation.png  
  part2/  
    [technical response files, if applicable]
```

Use of AI tools

Use of AI tools is permitted but not required for this assessment.

You may use AI tools to support or perform aspects of the log analysis. For example, AI may be used to help interpret the authentication-log format, suggest analysis questions, or identify possible patterns.

However, AI-generated analysis is not sufficient evidence on its own. All quantitative findings, patterns, tables, and visualisations used in your report must be verified using your submitted Python script against your assigned authentication-log extract.

If your group uses AI tools, you should include an AI-use declaration as an appendix to the report. The declaration should explain:

- which AI tool or tools were used
- what the tools were used for
- how the AI-supported outputs were checked, modified, rejected, or built upon
- which parts of the submitted work, if any, were substantially shaped by AI
- how the group ensured that the final claims were supported by Python-verified evidence

If no AI-use declaration is included, the course team will assume that the group is declaring that no AI tools were used. Regardless of whether AI tools are used, the group remains responsible for the accuracy, interpretation, and integrity of the submitted work.

Group workload record and Personal Assessment of Contribution

This is a group assessment. Groups may include a one-page workload record as an appendix to the report. This can take any form the group considers appropriate. For example, it may describe how work was divided across log analysis, Python development, visualisation, risk matrix development, Part 2 technical design, report writing, review, and coordination.

Groups may simply state: **“All members contributed equally.”**

If no workload record is included, the course team will assume that the group is declaring that all members contributed equally.

Each group member also has the option to submit a private Personal Assessment of Contribution via Moodle. In this statement, the student may divide 100 points across all members of the group, including themselves, to indicate their view of relative contribution.

If a student does not submit a Personal Assessment of Contribution, this will be interpreted as the student indicating that all members contributed equally.

Marking criteria

The mini-project is worth 20% of the overall course mark. Part 1 and Part 2 each contribute 10%.

Criterion	Marks
Part 1: Data-Driven Authentication Log Analysis	10
Part 2: Technical Defensive Response	10
Overall artefact quality	2

Part 1: Data-Driven Authentication Log Analysis.

Criterion	Marks	Description
Python analysis, reproducibility and reviewability	4	The script runs on the assigned authentication-log extract and can process comparable logs with the same general format. It extracts relevant fields, generates the submitted outputs, avoids unnecessary hard-coding, is clearly structured and commented, and includes a README that would allow a later review team to reproduce the evidence.
Data exploration, evidence and visualisation	3	The submission provides accurate summaries and at least two meaningful Python-generated visualisations, including top source IPs by failed authentication attempts. Evidence is clearly connected to claims in the report.
Security interpretation and risk prioritisation	2	The report interprets authentication patterns as possible threats or vulnerabilities, avoids overclaiming, and prioritises 3 to 5 risks or findings in relation to a clearly identified asset. Likelihood and impact are justified with reference to Python-verified evidence, the asset under consideration, rationale, and uncertainty.
Response recommendations and limitations	1	The report provides proportionate initial response recommendations and explains assumptions, missing evidence, and limitations.

Part 2: Technical Defensive Response. Part 2 marking criteria will assess the quality of the technical defensive response, including the connection to a Part 1 risk or finding, the technical appropriateness of the response, the justification of design decisions, consideration of limitations or trade-offs, and clarity of communication.

Overall artefact quality. Marks are awarded for the quality, coherence, professionalism, and usability of the submitted artefact as a whole. This includes clear organisation, appropriate presentation of figures and tables, consistency between the report and supporting files, clarity of explanation, completeness of the submission package, and the extent to which the materials could be understood, reviewed, and reused by a subsequent team.

Submission checklist

CHECKPOINT

Before submission, check that your package includes:

- ☐ report.pdf
- ☐ README.md
- ☐ analysis.py
- ☐ output folder containing generated tables and figures
- ☐ asset-focused risk matrix appendix
- ☐ AI-use declaration if AI tools were used
- ☐ workload record if the group wishes to include one
- ☐ any Part 2 files required by the technical response

Safe and ethical practice

This assessment uses provided authentication-log data for defensive analysis. You must not attempt to access, scan, attack, or interact with any live system as part of this assessment.

Your work should focus on analysing the provided dataset, interpreting evidence, and proposing defensive responses. Any technical recommendations should be framed as defensive measures intended to improve security, monitoring, investigation, or resilience.

Do not include instructions that would enable unauthorised access, exploitation, or evasion. Where you discuss attacker behaviour, do so only to support defensive reasoning and risk communication.

Plagiarism and Integrity

Students are required to familiarise themselves with the University of Glasgow Plagiarism & Academic Integrity Code. The code is available online at <https://www.gla.ac.uk/myglasgow/apg/policies/uniregs/regulations2025-26/feesandgeneral/studentssupportandconductmatters/reg32/>.

Student Learning Development (SLD) service provides a range of services to support students, including guidance on plagiarism as well as a dedicated Moodle with guidance on how to avoid it. The resources are available online at <https://www.gla.ac.uk/myglasgow/sld/>.

If you need advice about avoiding plagiarism or are unsure what is deemed as plagiarism, then immediately seek guidance from any of the following:

- Lecturer
- Course coordinator
- Dissertation supervisor
- Adviser of studies
- Student learning services

Submission Details

The submission window for the Mini-project opens on Monday the 20th of July 2026 and closes on Friday the 24th of July 2026 at 1630.

A nominated member of the team can make the group submission, and the submitted artefacts are associated with every other team member.

If you are unsure if your group has submitted artefacts, any team member can check what has been submitted, or is associated with them, by clicking on the submission window.

The nominated team member is expected to complete a **Submission Checklist** before the submission link becomes active.